8.2 More General Deterministic Thermostats

We have now seen that it is possible to construct stochastic differential equations that preserve the Gibbs distribution or to design deterministic perturbations of Hamiltonian dynamics, in an extended phase space, that do the same thing. The latter class of methods are not rigorously ergodic, but the dynamics based thermostatting technique adds a great deal of flexibility. For example it is easy to control the ensemble using dynamic perturbations introduced only in some special phase space directions. In this section, we show how to design distributional controls which blend deterministic dynamics with restricted stochastic perturbations, with, in some cases, provable ergodicity. We will present the methods of this section in some generality.

Let a system be provided with Hamiltonian (energy) function $H = H(z), z \in \mathbb{R}^m$, and microcanonical dynamics

$$\dot{z} = \boldsymbol{J}\nabla H, \tag{8.14}$$

where $\boldsymbol{J}$ is skew symmetric. In principle, $\boldsymbol{J}$ need not be constant, as long as it satisfies certain constraints, but as the extra generality is not needed for our current purposes, we restrict the case $\boldsymbol{J}$ constant. We describe dynamical and stochastic-dynamical techniques for computing averages with respect to a smooth density, for simplicity the Gibbsian one, $\rho_\beta \propto \exp(-\beta H)$.

We first consider methods that are based on perturbation of the conservative system (8.14), i.e.

$$\dot{z} = \boldsymbol{J}\nabla H + \boldsymbol{\Phi}(\boldsymbol{\xi}, z), \tag{8.15}$$

where $\boldsymbol{\xi} \in \mathbb{R}^m$ is controlled by the dynamical system

$$\frac{\mathrm{d}\boldsymbol{\xi}}{\mathrm{d}t} = \boldsymbol{g}(\boldsymbol{\xi}, z).$$

8.2.1 Continuity Equation

Given a deterministic thermostatting method, the density will evolve under the continuity equation,

$$\frac{\partial \rho}{\partial t} = -\nabla_z \cdot [\rho(\boldsymbol{J}\nabla_z \boldsymbol{H} + \boldsymbol{\Phi})] - \nabla_\xi \cdot (\rho \boldsymbol{g}(\boldsymbol{\xi}, z)) =: \mathcal{L}^\dagger \rho.$$

We have, using product rule,

$$\mathcal{L}^\dagger \rho = -(\nabla_z \rho) \cdot (\boldsymbol{J}\nabla_z H + \boldsymbol{\Phi}) - \rho \nabla_z \cdot (\boldsymbol{J}\nabla_z H + \boldsymbol{\Phi}) - (\nabla_\xi \rho) \cdot \boldsymbol{g} - \rho \nabla_\xi \cdot \boldsymbol{g}.$$

We divide this into four parts: $\mathcal{L}^\dagger \rho = W_1 + W_2 + W_3 + W_4$, where

$$W_1 = -(\nabla_z \rho) \cdot (\boldsymbol{J}\nabla_z H + \boldsymbol{\Phi}),\ W_2 = -\rho \nabla_z \cdot (\boldsymbol{J}\nabla_z H + \boldsymbol{\Phi}),$$
$$W_3 = -(\nabla_\xi \rho) \cdot \boldsymbol{g},\qquad\qquad W_4 = -\rho \nabla_\xi \cdot \boldsymbol{g}.$$

We seek perturbations of Hamiltonian dynamics with separable equilibria, i.e., such that $\rho = \rho_\beta(H(z))\hat{\rho}(\boldsymbol{\xi})$ is a stationary solution.

Taking this as an ansatz, the first term can be simplified as follows:

$$\begin{aligned} W_1 &= -\hat{\rho}(\boldsymbol{\xi})\rho'_\beta(H)\nabla_z H \cdot (\boldsymbol{J}\nabla_z H + \boldsymbol{\Phi}) \\ &= \beta\hat{\rho}(\boldsymbol{\xi})\rho_\beta(H)\nabla_z H \cdot \boldsymbol{\Phi} = \beta\rho\nabla_z H \cdot \boldsymbol{\Phi} \end{aligned}$$

as $\boldsymbol{J}$ is skew-symmetric.

Using the volume preservation of Hamiltonian systems, the second term satisfies

$$W_2 = -\rho\nabla_z \cdot (\boldsymbol{J}\nabla_z H + \boldsymbol{\Phi}) = -\rho\nabla_z \cdot \boldsymbol{\Phi},$$

hence

$$W_1 + W_2 = -\rho[\nabla_z - \beta\nabla_z H] \cdot \boldsymbol{\Phi}.$$

There are many possibilities for the distribution of $\boldsymbol{\xi}$, of course, but we will assume this is Gaussian for simplicity: $\hat{\rho}(\boldsymbol{\xi}) = \exp\left(-\beta \sum_{i=1}^m \mu_i \xi_i^2/2\right) = \exp(-\beta \boldsymbol{\xi}^T \boldsymbol{\mu}^{-1}\boldsymbol{\xi}/2)$ to get

$$W_3 = -(\nabla_\xi \rho) \cdot \boldsymbol{g}(\boldsymbol{\xi}, z) = \beta\rho\boldsymbol{\xi}^T \boldsymbol{\mu}^{-1}\boldsymbol{g}.$$

Finally, for an equilibrium solution we must have $W_1 + W_2 + W_3 + W_4 = 0$, so

$$-[\nabla_z - \beta\nabla_z H] \cdot \boldsymbol{\Phi} + \beta\boldsymbol{\xi}^T \boldsymbol{\mu}^{-1}\boldsymbol{g} - \nabla_\xi \cdot \boldsymbol{g} = 0.$$

This equation is the fundamental one for deterministic thermostats.

Example 8.1 To recover Nosé-Hoover, assume an even dimensional phase space, $z = (\boldsymbol{q}, \boldsymbol{p})$, $\mathrm{N_d}$ degrees of freedom, and $\xi \in \mathbb{R}$,

$$\boldsymbol{\Phi} = -\xi \begin{bmatrix} \boldsymbol{0} \\ \boldsymbol{p} \end{bmatrix}, \qquad g(z) = \mu^{-1}(\boldsymbol{p} \cdot \nabla_{\boldsymbol{p}} H - \mathrm{N_d k_B T}).$$

Example 8.2 Slightly more generally, let

$$\boldsymbol{\Phi} = \xi \boldsymbol{G}(z), \qquad g(\xi, z) = g(z) = \mu^{-1}[\nabla_z H - \mathrm{k_B T}\nabla_z] \cdot \boldsymbol{G}(z).$$